# Assessment Menu: Teacher Guide

The Similarity Bistro · design notes, grading and answer key

**Why students cannot just pick the easy ones**

1. **Points are capped by category.** Students must complete exactly 4 / 4 / 2 dishes, so they cannot fill the total with Appetizers.
2. **Chef's Specials carry 40% of the grade** and Entrées another 40%. A student who only masters the basics tops out at 20 points.
3. **Skill coverage rule.** Their 10 picks must include at least one dish from each skill group below, so every strand of the unit is assessed.
4. **Harder dishes reward process.** Each has a rubric that gives credit for correct steps, so a partly correct attempt at a harder problem earns more than a perfect easy one in the same time.

| Skill group | What it covers | Dishes |
|---|---|---|
| **SCALE** | Similar figures, scale factor and proportions | A1, A2, E1, S1 |
| **CRITERIA** | AA, SAS and SSS similarity | A3, A4, E2, S2 |
| **RATIOS** | Perimeter and area ratios | A5, E3, E6, S3 |
| **APPLY** | Indirect measurement and triangle proportionality | A6, E4, E5, S4 |

| If a student only does… | Maximum |
|---|---|
| Only Appetizers (4 × 5) | 20 |
| Appetizers + Entrées (4 × 5 + 4 × 10) | 60 |
| Full meal (adds 2 Chef's Specials) | 100 |

**Grading and options**

- Grade only the marked dishes using the rubrics on the next pages (partial credit is built in). Total possible: **100**. If a student marks more than the required number, grade the first in each category.
- Check the skill rule on the order form before totaling. If a skill group is missing, apply your own policy (for example, the dish that broke the rule scores 0, or the student may swap one dish).
- Optional "second helping": allow one extra dish per category that replaces the student's lowest score in that category.
- Look at *which* dishes students chose. Students who select only the easiest dish in each skill group may need a conference.

| Student | Appetizers (of 20) | Entrées (of 40) | Specials (of 40) | Skills OK? | Total |
|---|---|---|---|---|---|
| | | | | | |
| | | | | | |

# Answer Key: Appetizers (5 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

| Dish | Problem | Solution | Rubric |
|---|---|---|---|
| **A1** SCALE | Rectangle A is 4 by 6. Rectangle B is similar to A, and its shorter side is 10. Find the scale factor from A to B and the longer side of B. | **Answer:** Scale factor $\frac{5}{2}$; longer side = 15<br>Scale factor = $\frac{10}{4} = \frac{5}{2}$.<br>Longer side = $6 \times \frac{5}{2} = 15$. | 2 pts: Scale factor $\frac{5}{2}$ (or 2.5)<br>3 pts: Longer side 15 |
| **A2** SCALE | $\triangle ABC \sim \triangle DEF$ with AB = 6, DE = 9, and BC = 10. Write a proportion and find EF. | **Answer:** $\frac{6}{9} = \frac{10}{EF}$, so EF = 15<br>Corresponding sides are proportional: $\frac{6}{9} = \frac{10}{EF}$<br>Cross-multiply: $6 \cdot EF = 90$, so EF = 15. | 2 pts: Correct proportion with matching sides<br>3 pts: Solves EF = 15 |
| **A3** CRITERIA | Triangle 1 has angles 50° and 60°. Triangle 2 has angles 60° and 70°. Are the triangles similar? Show the third angles and name the criterion. | **Answer:** Yes, by AA: both have angles 50°, 60°, 70°.<br>Third angle of triangle 1: $180° - 50° - 60° = 70°$.<br>Third angle of triangle 2: $180° - 60° - 70° = 50°$.<br>Both have 50°, 60°, 70°, so they are similar by **AA**. | 2 pts: Finds both third angles (70° and 50°)<br>1 pts: Says similar<br>2 pts: Names AA |
| **A4** CRITERIA | $\triangle ABC$ has AB = 4, AC = 6, $\angle A = 50°$. $\triangle DEF$ has DE = 10, DF = 15, $\angle D = 50°$. Are they similar? Show the ratios and name the criterion. | **Answer:** Yes, by SAS~: $\frac{4}{10} = \frac{6}{15} = \frac{2}{5}$ with $\angle A = \angle D$.<br>$\frac{4}{10} = \frac{2}{5}$ and $\frac{6}{15} = \frac{2}{5}$, so the sides around $\angle A$ and $\angle D$ are proportional.<br>The included angles are equal (50°), so the triangles are similar by **SAS~**. | 2 pts: Both ratios equal $\frac{2}{5}$<br>1 pts: Notes the included angles are equal<br>2 pts: Names SAS similarity |
| **A5** RATIOS | Two similar triangles have a scale factor of 3 : 5. The smaller one has perimeter 18. Find the perimeter of the larger one. | **Answer:** 30<br>Perimeters have the same ratio as the sides: $\frac{18}{P} = \frac{3}{5}$<br>3P = 90, so P = 30. | 2 pts: Uses the ratio 3 : 5 for perimeters<br>3 pts: Perimeter 30 |
| **A6** APPLY | A 2 m stick casts a 3 m shadow. At the same moment a tree casts a 12 m shadow. Write a proportion and find the tree's height. | **Answer:** $\frac{2}{3} = \frac{h}{12}$, so h = 8 m<br>The sun makes similar triangles: $\frac{2}{3} = \frac{h}{12}$<br>3h = 24, so h = 8 m. | 2 pts: Correct proportion (height to shadow)<br>2 pts: Solves h = 8<br>1 pts: Includes units |

# Answer Key: Entrées (10 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

| Dish | Problem | Solution | Rubric |
|---|---|---|---|
| **E1** SCALE | △ABC ~ △DEF. AB = 8, DE = 12, BC = x + 2, and EF = 3x − 3. Find x, BC, and EF. | **Answer:** x = 4, BC = 6, EF = 9<br>$\frac{8}{12} = \frac{x+2}{3x-3}$<br>Cross-multiply: 8(3x − 3) = 12(x + 2), so 24x − 24 = 12x + 24.<br>12x = 48, so x = 4. BC = 6 and EF = 9. | 3 pts: Correct proportion with matching sides<br>3 pts: Cross-multiplies and distributes correctly<br>2 pts: x = 4<br>2 pts: BC = 6 and EF = 9 |
| **E2** CRITERIA | △ABC has sides 6, 8, 10. △DEF has sides 9, 12, 15. △GHI has sides 9, 12, 16. Is △ABC similar to △DEF? To △GHI? Show the ratios and name the criterion. | **Answer:** Similar to △DEF (SSS~, ratio $\frac{2}{3}$); not similar to △GHI.<br>DEF: $\frac{6}{9} = \frac{8}{12} = \frac{10}{15} = \frac{2}{3}$, so similar by **SSS~**.<br>GHI: $\frac{6}{9} = \frac{8}{12} = \frac{2}{3}$ but $\frac{10}{16} = \frac{5}{8}$, so the ratios are not all equal: not similar. | 6 pts: Matches sides in order; DEF: all ratios $\frac{2}{3}$, names SSS~<br>4 pts: GHI: finds $\frac{5}{8} \neq \frac{2}{3}$, concludes not similar |
| **E3** RATIOS | Two similar triangles have areas 20 and 45. Find the ratio of their corresponding sides. If the smaller triangle has perimeter 14, find the perimeter of the larger. | **Answer:** Sides 2 : 3; larger perimeter = 21<br>Area ratio = $\frac{20}{45} = \frac{4}{9}$.<br>Side ratio: $\sqrt{\frac{4}{9}} = \frac{2}{3}$.<br>Perimeters use the side ratio: $\frac{14}{P} = \frac{2}{3}$, so P = 21. | 3 pts: Area ratio $\frac{4}{9}$<br>3 pts: Takes the square root to get 2 : 3<br>4 pts: Perimeter 21 using the side ratio |
| **E4** APPLY | In △ABC, D is on AB and E is on AC with DE ∥ BC. AD = 6, DB = 4, and AE = 9. Find EC. If DE = 12, find BC. | **Answer:** EC = 6; BC = 20<br>Triangle proportionality: $\frac{AD}{DB} = \frac{AE}{EC}$, so $\frac{6}{4} = \frac{9}{EC}$ and EC = 6.<br>△ADE ~ △ABC (AA) with AB = 6 + 4 = 10: $\frac{6}{10} = \frac{12}{BC}$, so BC = 20. | 3 pts: Proportion $\frac{6}{4} = \frac{9}{EC}$<br>2 pts: EC = 6<br>3 pts: Uses AB = 10 (not DB = 4) in △ADE ~ △ABC<br>2 pts: BC = 20 |
| **E5** APPLY | A mirror lies flat on the ground 6 m from the base of a flagpole. Maya stands 1.5 m past the mirror, on the same line, and just sees the top of the pole in it. Her eyes are 1.6 m high. How tall is the pole? | **Answer:** 6.4 m<br>Right angles and equal angles of reflection make the triangles similar (**AA**).<br>Pole to mirror matches eye height to Maya's distance: $\frac{h}{6} = \frac{1.6}{1.5}$<br>$h = \frac{6 \times 1.6}{1.5} = 6.4$ m. | 3 pts: Explains why the triangles are similar (right angles and equal reflection angles)<br>3 pts: Correct proportion with matching sides<br>2 pts: h = 6.4<br>2 pts: Units |
| **E6** RATIOS | Two similar triangles have sides in the ratio 2 : 3. The smaller has area 12. Dan finds the larger area as $12 \times \frac{3}{2} = 18$. Find his mistake, explain it, and give the correct area. | **Answer:** He used the side ratio for area. The area ratio is 4 : 9, so the larger area is 27.<br>Areas are in the ratio of the **squares** of the sides: $2^2 : 3^2 = 4 : 9$.<br>$\frac{12}{A} = \frac{4}{9}$, so A = 27. | 4 pts: Identifies the mistake: used the side ratio instead of the area ratio<br>3 pts: Explains that area ratio = (side ratio)$^2$ = 4 : 9<br>3 pts: Correct area 27 |

# Answer Key: Chef’s Specials (20 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

| Dish | Problem | Solution | Rubric |
|---|---|---|---|
| **S1** SCALE | Quadrilaterals ABCD ~ WXYZ (in that order). AB = 6, WX = 9, BC = 2x + 2, XY = 4x − 1. (a) Find x and the scale factor from ABCD to WXYZ. (b) WXYZ has perimeter 60; find the perimeter of ABCD. (c) ABCD has area 48; find the area of WXYZ. | **Answer:** (a) x = 4, scale factor $\frac{3}{2}$ (b) 40 (c) 108<br>(a) $\frac{6}{9} = \frac{2x+2}{4x-1}$, so 3(2x + 2) = 2(4x − 1): 6x + 6 = 8x − 2, so x = 4. BC = 10 and XY = 15. Scale factor ABCD to WXYZ = $\frac{9}{6} = \frac{3}{2}$.<br>(b) $\frac{P}{60} = \frac{2}{3}$, so P = 40.<br>(c) Area ratio is $\frac{9}{4}$: $\frac{A}{48} = \frac{9}{4}$, so A = 108. | 6 pts: (a) Correct proportion (2), solves x = 4 (4)<br>2 pts: (a) Scale factor $\frac{3}{2}$ (direction ABCD to WXYZ)<br>12 pts: (b) Perimeter 40 using the side ratio (5); (c) squares the ratio to get $\frac{9}{4}$ (4), area 108 (3) |
| **S2** CRITERIA | In △ABC, D is on AB and E is on AC with AD = 4, AB = 10, AE = 6, AC = 15. (a) Show △ADE ~ △ABC and name the criterion. (b) Explain why DE ∥ BC. (c) If BC = 20, find DE. | **Answer:** (a) SAS~ (ratio $\frac{2}{5}$) (b) corresponding angles equal (c) DE = 8<br>(a) $\frac{AD}{AB} = \frac{4}{10} = \frac{2}{5}$ and $\frac{AE}{AC} = \frac{6}{15} = \frac{2}{5}$. ∠A is shared, so △ADE ~ △ABC by **SAS~**.<br>(b) Similar triangles have equal corresponding angles, so ∠ADE = ∠ABC. These are corresponding angles for lines DE and BC with transversal AB, so DE ∥ BC.<br>(c) $\frac{DE}{20} = \frac{2}{5}$, so DE = 8. | 8 pts: (a) Both ratios $\frac{2}{5}$ (4), shared ∠A (2), names SAS similarity (2)<br>6 pts: (b) ∠ADE = ∠ABC (3); converse of corresponding angles gives parallel lines (3)<br>6 pts: (c) Proportion $\frac{DE}{20} = \frac{2}{5}$ (3); DE = 8 (3) |
| **S3** RATIOS | Two similar triangular sails: the larger has area 144 ft² and perimeter 48 ft; the smaller has perimeter 32 ft. (a) Find the scale factor from large to small. (b) Find the smaller area. (c) A side of the smaller is 8 ft; find the matching side of the larger. (d) Why does area change by the square? | **Answer:** (a) $\frac{2}{3}$ (b) 64 ft² (c) 12 ft (d) area is length × length, so both dimensions scale<br>(a) $\frac{32}{48} = \frac{2}{3}$.<br>(b) Area ratio = $\frac{4}{9}$: 144 × $\frac{4}{9}$ = 64 ft².<br>(c) $\frac{8}{s} = \frac{2}{3}$, so s = 12 ft.<br>(d) Both dimensions are multiplied by the scale factor, so area is multiplied by (scale factor)$^2$. | 4 pts: (a) Scale factor $\frac{2}{3}$<br>6 pts: (b) Squares to get $\frac{4}{9}$ (3); area 64 (3)<br>5 pts: (c) Proportion (2); 12 ft (3)<br>5 pts: (d) Explains that both dimensions scale, so area scales by the square |
| **S4** APPLY | A lamp is H ft above the ground on a vertical pole. Nia, 6 ft tall, stands 20 ft from the base of the pole and her shadow is 10 ft long. (a) Find H and explain why the triangles are similar. (b) She walks to 30 ft from the pole. How long is her shadow now? | **Answer:** (a) H = 18 ft (b) 15 ft<br>(a) Pole and Nia are both vertical and share the angle at the shadow tip: similar (**AA**). Bases are 30 and 10: $\frac{H}{30} = \frac{6}{10}$, so H = 18.<br>(b) $\frac{18}{30+s} = \frac{6}{s}$: 18s = 180 + 6s, so s = 15 ft. | 4 pts: (a) Explains similarity (vertical pole and Nia, shared angle)<br>6 pts: (a) Proportion using base 30 (4); H = 18 (2)<br>10 pts: (b) Proportion $\frac{18}{30+s} = \frac{6}{s}$ (6); s = 15 ft (4) |